

CS & IT ENGINEERING



Computer Network

IPv4 Address

Lecture No. - 10



By - Abhishek Sir



Recap of Previous Lecture



Topic

Routing Table

Topic

CIDR



Topics to be Covered



Topic

CIDR

Topic

IP Configuration

ABOUT ME



Hello, I'm **Abhishek**

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[MCQ]

[GATE-CS-2004] [2 Mark]



#Q. The routing table of a router shown below :

Destination	Subnet Mask	Interface
128.75.43.0	255.255.255.0	Eth0
128.75.43.0	255.255.255.128	Eth1
192.12.17.5	255.255.255.255	Eth3
Default 0.0.0.0	0.0.0.0	Eth2

→ PKT₁

→ PKT₂

On which interfaces will the router forward packets addressed to destinations 128.75.43.16 and 192.12.17.10 respectively ?

✓ A Eth1 and Eth2

~~C~~ Eth0 and Eth3

~~B~~ Eth0 and Eth2

~~D~~ Eth1 and Eth3

Ans: A

128 . 75 . 43 . 16
255 . 255 . 255 . 255

128 . 75 . 43 . 16

Not matched

128 . 75 . 43 . 16
255 . 255 . 255 . 128

128 . 75 . 43 . 0

192 . 12 . 17 . 10
255 . 255 . 255 . 255

192 . 12 . 17 . 10

#Q. The forwarding table of a router is shown below.

Subnet Number	Subnet Mask	Interface ID
200.150.0.0	255.255.0.0	1
200.150.64.0	255.255.224.0	2
200.150.68.0	255.255.255.0	3
200.150.68.64	255.255.255.240	4
Default		0

A packet addressed to a destination address [200.150.68.118] arrives at the router. It will be forwarded to the interface with ID _____.

Ans: 3

200.150. 68.118

255.255.255. ~~224~~240

200.150. 68. ~~224~~112

Not matched ↑

118 → 01110110

240 → 11110000

112 ← 01110000

200.150. 68.118

255.255.255. 0

200.150. 68. 0

Matched

[MCQ]

[GATE-2025][2 Mark]



#Q. A packet with the destination IP address 145.36.109.70 arrives at a router whose routing table is shown. Which interface will the packet be forwarded to?

Subnet	Subnet Mask (in CIDR notation)	Interface
145.36.0.0	/16	E1
145.36.128.0	/17	E2
<u>145.36.64.0</u>	/18	E3 ✓
<u>145.36.225.0</u>	/24	E4 ✗
Default	--	E5

- ✓ A E3
- B E1
- C E2
- D E5

Ans: A

145. 36. 109. 70

255. 255. 255. 0

145. 36. 109. 0

Not matched

145. 36. 109. 70

255. 255. 192. 0

145. 36. 64. 0

↑

Matched

109 → 01101101

192 → 11000000

64 ← 01000000

[MCQ]

[GATE-2020][2 Mark]



#Q. An organization requires a range of IP address to assign one to each of its 1500 computers. The organization has approached an Internet Service Provider (ISP) for this task. The ISP uses CIDR and serves the requests from the available IP address space "202.61.0.0/17". The ISP wants to assign an address space to the organization which will minimize the number of routing entries in the ISP's router using route aggregation. Which of the following address spaces are potential candidates from which the ISP can allot any one of the organization?

~~I. 202.61.84.0/21~~

84 → 01010100

144 → 10010000

✓ II. 202.61.104.0/21

104 → 01101000

✓ III. 202.61.64.0/21

~~IV. 202.61.144.0/21~~

64 → 01000000

~~A~~ I and II only

✓ B II and III only

C III and IV only

D I and IV only

Ans: B

Available IP Address space = 202.61.0.0 / 17

202.61.0 0 0 0 0 0 0 0 0 . 0 0 0 0 0 0 0 0 / 17
 17 bit prefix 15 bit host ID

Number of hosts = 1500
 [for IP address requested]

Number of bits for host id in allocated chunk = $\lceil \log_2(\text{Number of hosts}) \rceil \text{ bits}$
 $= \lceil \log_2(1500) \rceil \text{ bits} = 11 \text{ bits}$

Allocated IP Address space =
 202.61.0 P₁ P₂ P₃ P₄ 0 0 0 . 0 0 0 0 0 0 0 0 / 21
 21 bit prefix 11 bit Host ID

[MCQ]

[GATE-2012][2 Mark]



#Q. An Internet Service Provider (ISP) has the following chunk of CIDR-based IP addresses available with it : "245 . 248 . 128 . 0 / 20". The ISP wants to give half of this chunk of addresses to Organization A, and a quarter to Organization B, while retaining the remaining with itself. Which of the following is a valid allocation of addresses to A and B? (respectively)

- ☒ A 245 . 248 . 136 . 0 / 21 and 245 . 248 . 128 . 0 / 22
- ☐ B 245 . 248 . 128 . 0 / 21 and 245 . 248 . 128 . 0 / 22
- ☐ C 245 . 248 . 132 . 0 / 22 and 245 . 248 . 132 . 0 / 21
- ☐ D 245 . 248 . 136 . 0 / 24 and 245 . 248 . 132 . 0 / 21

Ans: A

CIDR based available IP address space : 245.248.128.0 / 20

245.248.1000 0000 . 000000000000 / 20
 20 bit prefix 12 bit Host ID

Org.-A Network Address : 245.248.128.0/21

245.248.1000 0 0000 . 000000000000 / 21
 21 bit prefix 11 bit Host ID

Org.-B Network Address : 245.248.136.0/22

245.248.1000 1 000 . 000000000000 / 22
 22 bit prefix 10 bit Host ID

ISP available Address Space : 245.248.140.0/22

245.248.1000 1 00 . 000000000000 / 22
 22 bit prefix 10 bit Host ID

CIDR based available IP address space : 245 . 248 . 128 . 0 / 20

245 . 248 . 1000 _____ / 20
 20 bit prefix 12 bit HostID

Org.-1 Network Address : 245 . 248 . 136 . 0 / 21

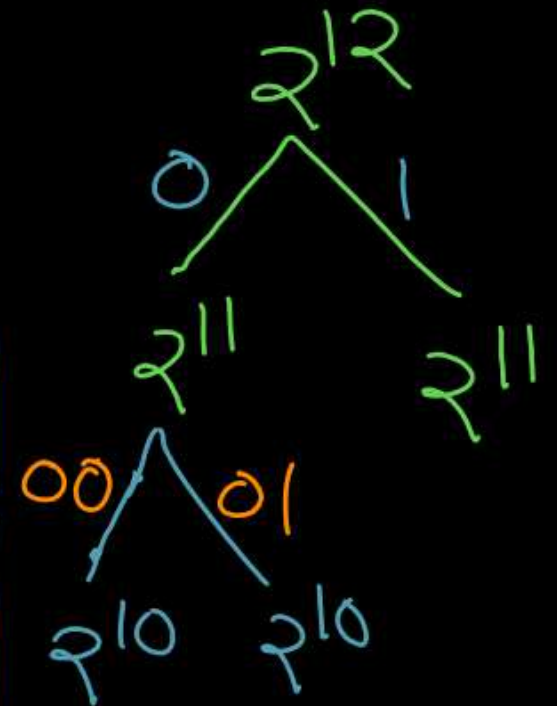
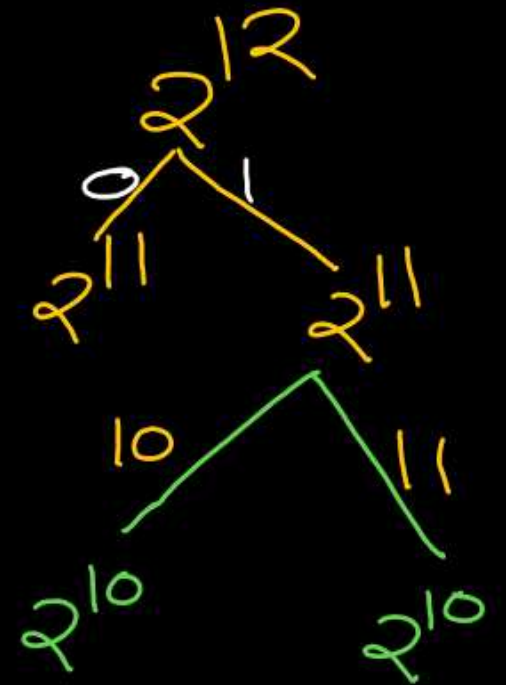
245 . 248 . 1000 1 000 . 000000000000 / 21
 21 bit prefix 11 bits HostID

Org.-2 Network Address : 245 . 248 . 128 . 0 / 22

245 . 248 . 1000 0 0 00 . 000000000000 / 22
 22 bit prefix 10 bit HostID

ISP available Address Space : 245 . 248 . 132 . 0 / 22

245 . 248 . 1000 0 1 00 . 000000000000 / 22
 22 bit prefix 10 bit HostID



Org.-1 Net. Add. : 245 . 248 . 128.0/21
Org.-2 Net. Add. : 245 . 248 . 136.0/22
ISP Remaining : 245 . 248 . 140.0/22

✓
Org.-1 Net. Add. : 245 . 248 . 136.0/21
Org.-2 Net. Add. : 245 . 248 . 128.0/22
ISP Remaining : 245 . 248 . 132.0/22

Org.-1 Net. Add. : 245 . 248 . 128.0/21
Org.-2 Net. Add. : 245 . 248 . 140.0/22
ISP Remaining : 245 . 248 . 136.0/22

Org.-1 Net. Add. : 245 . 248 . 136.0/21
Org.-2 Net. Add. : 245 . 248 . 132.0/22
ISP Remaining : 245 . 248 . 128.0/22

Example :-

[MCQ]



#Q. Which one of the following CIDR prefixes exactly represents the range of IP addresses 192.168.0.0 to 192.168.255.255?

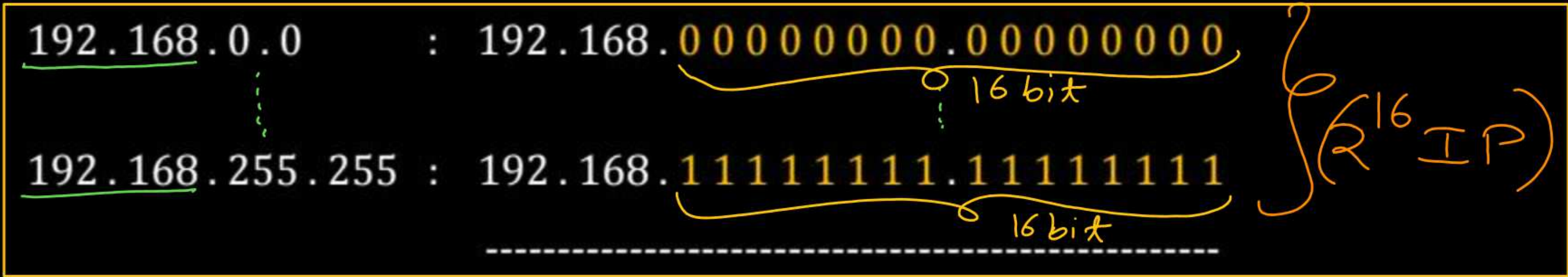
☒ A 192.168.0.0 / 16

☐ B 192.168.0.0 / 17

☐ C 192.168.0.0 / 23

☐ D 192.168.0.0 / 24

Ans: A



CIDR Prefixes : 192.168. 00000000.00000000 / 16
 Network Address : 16 bit prefix 16 bit Host ID

192.168.0.0/16

[MCQ] H.W.

11SC

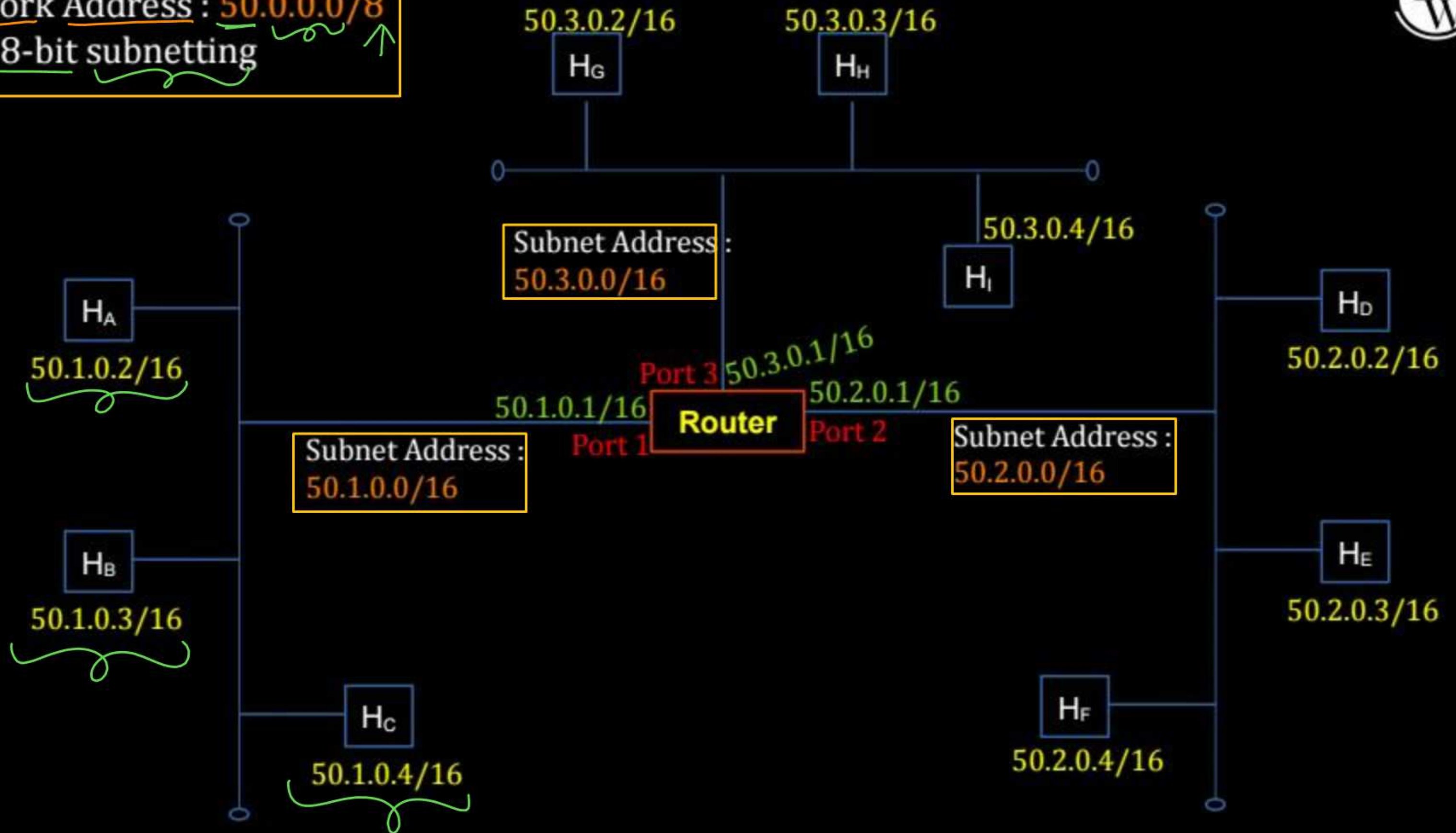
[GATE-2024][2 Mark]



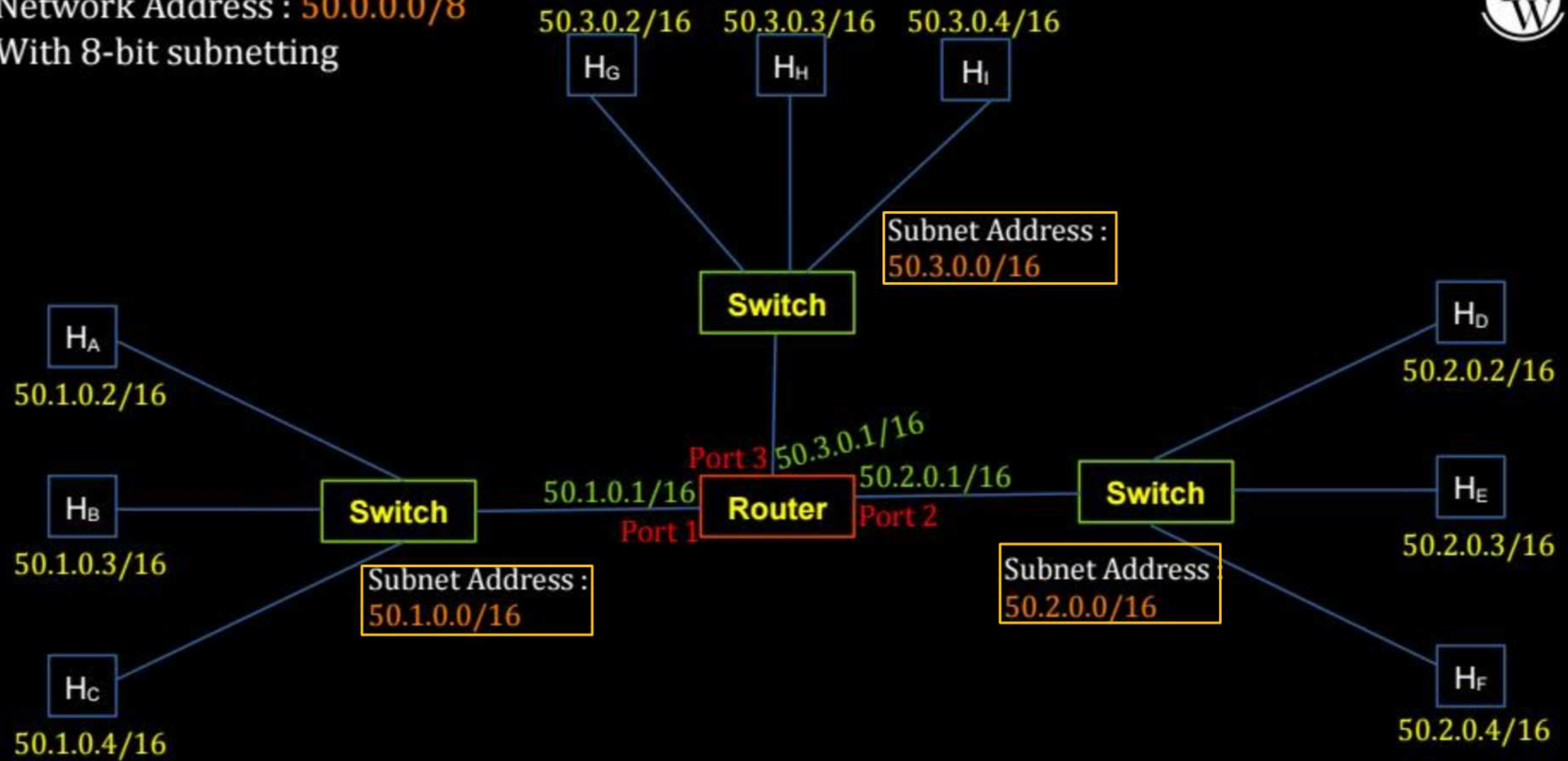
#Q. Which one of the following CIDR prefixes exactly represents the range of IP addresses 10.12.2.0 to 10.12.3.255?

- ☐ A 10.12.2.0/23
- ☐ B 10.12.2.0/24
- ☐ C 10.12.0.0/22
- ☐ D 10.12.2.0/22

Network Address : 50.0.0.0/8
 With 8-bit subnetting



Network Address : 50.0.0.0/8
With 8-bit subnetting



Host A, IPv4 Configuration :-

IPv4 Address : 50.1.0.2
 Subnet Mask : 16 or 255.255.0.0
 Default Gateway : 50.1.0.1

Router forwarding table

Subnet Address	Interface ID	Next Hop
50.1.0.0 / 16	1	On Link
50.2.0.0 / 16	2	On Link
50.3.0.0 / 16	3	On Link

CASE I:

IP Packet

Source IP Address : 50.1.0.2 H_A

Destination IP Address : 50.1.0.3 H_B

$$\begin{array}{r} 50.1.0.2 \\ 255.255.0.0 \\ \hline 50.1.0.0 \end{array}$$

$$\begin{array}{r} 50.1.0.3 \\ 255.255.0.0 \\ \hline 50.1.0.0 \end{array}$$

- Host A (source host) finds destination host IP (Host B) belongs to same subnet. ✓
- Host A uses ARP Protocol to find MAC Address of destination host (H_B)
- Host A send frame to Host B
[The frame encapsulates the IP datagram]

Source MAC Address : Host A MAC Address

Destination MAC Address : Host B MAC Address

Frame

[MCQ] H.W.

[GATE-CS-2006] [2 Mark]



#Q. Two computers C1 and C2 are configured as follows. C1 has IP address "203 . 197 . 2 . 53" and netmask "255 . 255 . 128 . 0". C2 has IP address "203 . 197 . 75 . 201" and netmask "255 . 255 . 192 . 0". Which one of the following statements is true ?

- A C1 and C2 both assume they are on the same network
- B C2 assumes C1 is on same network, but C1 assumes C2 is on a different network
- C C1 assumes C2 is on same network, but C2 assumes C1 is on a different network
- D C1 and C2 both assume they are on different networks



2 mins Summary



Topic

CIDR ✓

Topic

IP Configuration



THANK - YOU

